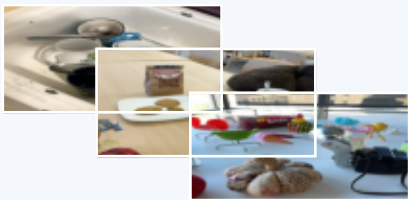


CTF-GS: Compact Foundation-Feature Gaussian Memory

Multiview evidence is compressed into a compact 3D memory, then query-conditioned feature/support reconstruction serves 2D and 3D open-vocabulary tasks.

A. Multiview Evidence



posed RGB views + 3DGS geometry

Frame-wise foundation features

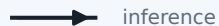
RADIO / SigLIP2 / DINO / SAM

Multiview primitive registration

visibility, contribution, confidence

Training constraints

feature, support, boundary, topology



inference

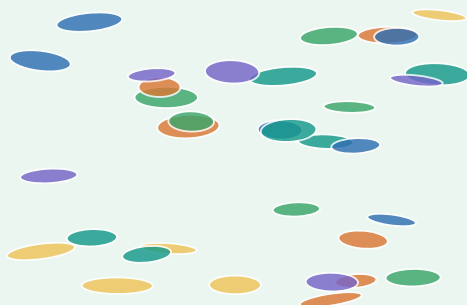


training only

B. Compact Scene Memory

Compact Foundation-Feature Memory

stored once per scene; reconstruct features on demand



compact code z_i + spatial context $h(x)$

not explicit 1280-D feature storage

compress

training only

C. Query-Conditioned Tasks

LERF rendered-view OVS

dense score map -> 2D mask



reconstruct

LERF direct 3D OVS

primitive scores -> selected mask



calibrate

ScanNet point query

open-vocabulary points



query

Frozen-head probes

SigLIP2 / SAM / DINO probes



probe